MIL-P-49139(EL)

22 May 1978

MILITARY SPECIFICATION

POWER SUPPLY PP-6148( )/U

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of power supply designated Power Supply PP-6148( )/U for furnishing 0-10 amperes at 14/28 vdc, from nominal 115/230 vac, 50, 60, or 400 Hz, single phase power sources. (See 6.1 and 6.3)

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Electronics Research and Development Command, ATTN: DELET-PE, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6130

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SPECIFICATI(

Federal

NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-78	Strapping, Steel, and Seals
PPP-B-5	Boxes, Wood, Wirebound
PPP-B-6	Boxes, Wood, Cleated Plywood
PPP-B-6	Boxes, Wood, Nailed and Lock-Corner
PPP-B-6	Boxes, Shipping, Fiberboard
PPP-B-6	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-3	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-7	Strapping, Nonmetallic (and Connectors.)
PPP-T-9	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging
MIL-C-39019	Circuit Breakers, Magnetic, Low-Power, Sealed, Trip Free, General Specification for
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-P-55110	Printed Wiring Boards
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Equipment
MIL-STD-147	Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-242	Electronic Equipment Parts, Selected Standards (Forward and Contents)
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipments
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

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DRAWINGS

Electronics Command

DL-SM-B-595320 Power Supply PP-6148( )/U

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification, and Drawing DL-SM-B-595320.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 4.4)

3.4 Printed wiring boards. Printed wiring boards used in the construction of printed wiring assemblies, shall be a product which has passed the requirements of MIL-P-55110, MIL-STD-275 and the master drawing. However, those parts of MIL-STD-275 which are in conflict with the applicable printed wiring board drawings and which require redesign of the printed wiring boards shall not apply. Conflicts between master drawings and MIL-STD-275 need not be submitted as required by MIL-P-55110. (See 3.3 and 4.4)

3.4.1 Protective coating. Printed wiring assemblies shall be protected on both sides of the printed wiring assembly with an epoxy-type conformal coating to protect the pattern against moisture, dust and other contaminants. The coating shall be Class A, Type ER or polyurethane, Type PUR in accordance with MIL-I-46058. The thickness of the coating shall be 0.0015 ±0.0005 inches. For areas where protective coating cannot be applied without impairment of electrical or physical parameters, the contractor shall coordinate with the contracting officer's technical representative for procedures for masking parts before applying the coating. Hermetic sealing of the assembly in a dry atmosphere shall be acceptable in place of conformal coating. (See 4.4)

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3.5 Marking. The method of marking the components shall conform to the requirements of MIL-STD-130. Front panel marking shall be Group 1 as described in MIL-M-13231. (See 4.4)

3.5.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.2 Serial number. Each power supply shall be serial numbered.

3.6 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as hinges, screws, etc., and necessary touch up after mounting. The final paint film on Type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 4.4 and 6.6)

3.7 Electrical requirements.

3.7.1 Input voltage. The power supply shall deliver the output required by this specification (see 4.7.1) from the following single phase input sources:

Volts ac (±10%)	Frequency (Hz)
115/230	50 to 400

3.7.2 Output voltage. The voltage across the output of the power supply shall be nominal 14 vdc (±1.5 percent), or 28 vdc (±1.0 percent) over the load range of 0-10 amperes. The output voltage shall also be adjustable over the range of 12-16 vdc and 24-32 vdc by an external locking type control. (See 4.7.1 and 4.7.1.1)

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3.7.3 Ripple. The rms ripple voltage shall not exceed 1.0 percent at any output condition from no-load to full-load. The peak to peak ripple voltage including internal switching spikes and other noise components shall not exceed ±2 percent at any output condition from no-load to full-load. The ripple voltage includes voltage perturbations within the frequency range of 20 Hz to 10 MHz. (See 4.7.2)

3.7.4 Recovery time. The output voltage of the power supply shall recover to 14/28 vdc (±1 percent) within 25 milliseconds when the load is switched from no-load to full-load and from full-load to no-load. The maximum instantaneous overshoot voltage under these conditions and for any turn on or turn off condition of the power supply shall not exceed 20 percent of the nominal voltage 14/28 vdc. (See 4.7.3)

3.7.5 Overload protection. The power supply shall be protected from physical and electrical damage from any overload conditions (including a direct short circuit) by the use of circuit breakers in the "ac input" circuit and the "dc output" circuit, and by automatic current limiting circuitry. The ac input circuit breaker shall be a double pole. The two poles shall each have dual ratings. The entire circuit breaker shall act as the "ON-OFF" switch. The "dc output" circuit breaker shall be a single pole rated at 10 amps and set for instantaneous tripping at 14 amps. Both circuit breakers shall be hydraulic-magnetic, sub-miniature, single-hole mounting, toggle type, and shall meet the requirements of MIL-STD-242 and MIL-C-39019. (See 4.7.4)

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3.7.6 DC output for battery charging.

3.7.6.1 Charging method, nickel cadmium batteries. Modified constant potential charging shall be employed. Current limiting setting shall be variable between 0.5 and 8.0 amperes and adjustable voltage capability provided over the range of 14 to 16 vdc or 28 to 32 vdc. (See 4.7.5.1)

3.7.7 Current limiting. Automatic control circuitry shall limit the output current to a maximum value of 10 ±0.5 amperes to protect the power supply from overload and automatically restore the output when the overload is removed. (See 4.7.4)

3.7.8 Protection.

3.7.8.1 Battery discharge. The power supply when used as a battery charger shall include provisions to prohibit the discharge of the battery being charged under the conditions of:

a. Loss of power input to the charger. (See 4.7.6)

b. Battery emf in excess of dc output voltage. (See 4.7.6)

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3.7.8.2 General. The power flow conversion circuitry and feedback control circuitry of the charger shall be fully protected under operational and abnormal conditions which include the following:

a. Short circuit of dc output. (See 4.7.4)

b. Reverse polarity of the battery at the output. (See 4.7.6)

c. Progressive and prolonged output overloads. (See 4.7.4)

d. Over voltage line surges and transient voltage spikes. (See 4.7.10)

e. Under voltage of the input voltage source below the minimum specified. (See 4.7.10)

The equipment shall automatically resume normal operation when any of the above faults are removed.

3.7.9 Efficiency. The overall efficiency of the power supply shall not be less than 65 percent. (See 4.7.7)

3.7.10 Stability (see 4.7.8). Output voltage drift at a constant load and ambient temperature shall not exceed 0.5 percent over a period of 24 hours.

3.7.11 Temperature coefficient (See 4.7.9). Change in output voltage as a function of ambient temperature shall not exceed 0.02 percent per degree F over the entire specified operating temperature range.

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3.8 Service conditions. Unless otherwise specified, prior to, during, and after testing as required for each of the following service conditions, the power supply shall conform to the operational test of 4.7.1.1. After testing, the power supply shall meet full specification performance for these measurements.

3.8.1 Temperature.

3.8.1.1 High. During testing as specified in 4.8.1.1, the power supply shall meet the following requirements:

Paragraph	Measurement
3.7.2	4.7.1
3.7.3	4.7.2
3.7.4	4.7.3
3.7.5	4.7.4
3.7.7	4.7.4

After testing, the power supply shall meet full specification performance for these measurements.

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3.8.1.2 Low. During testing as specified in 4.8.1.2, the power supply shall meet the following requirements:

Paragraph	Measurements
3.7.2	4.7.1
3.7.3	4.7.2
3.7.4	4.7.3
3.7.5	4.7.4
3.7.7	4.7.4

After testing the power supply shall meet full specification performance for these measurements.

3.8.2 Humidity. After testing as specified in 4.8.2, the power supply shall meet the following requirements:

Paragraph	Measurement
3.7.2	4.7.1
3.7.3	4.7.2
3.7.4	4.7.3

After testing, the power supply shall meet full specification performance for these measurements.

3.8.3 Altitude. The power supply shall meet full specification performance for the following measurements both during and after testing as specified in 4.8.3:

Paragraph	Measurement
3.7.2	4.7.1
3.7.3	4.7.2
3.7.4	4.7.3

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3.8.4 Orientation. When tested as specified in 4.8.4, the power supply shall meet full specification performance for the following measurement:

Paragraph	Measurement
3.7.2	4.7.1.1

3.8.5 Leakage (immersion). The power supply shall show no evidence of leakage after it is tested as specified in 4.8.5.

3.8.6 Vibration, bounce, shock (drop).

3.8.6.1 Vibration, internal. When tested as specified in 4.8.6, the amplitude of vibration of any part, subassembly, or structural member of the power supply shall not exceed twice the amplitude of the vibration applied to the power supply at any frequency between 10 and 55 Hz.

3.8.6.2 Bounce. The power supply as prepared for field use, shall withstand the vibrations normally induced during transportation as loose cargo. After testing as specified in 4.8.7 the equipment shall meet specification performance for the following measurements:

Paragraph	Measurement
3.7.2	4.7.1
3.7.3	4.7.2
3.7.4	4.7.3

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3.8.6.3 Shock (drop). The power supply as prepared for field use, shall withstand the shocks normally induced by loading and unloading of equipment. After testing as specified in 4.8.8 the equipment shall meet the specification performance for the following measurements:

Paragraph	Measurements
3.7.2	4.7.1
3.7.3	4.7.2
3.7.4	4.7.3

3.8.7 Fungus (See 4.8.9). After the fungus test of 4.8.9, the power supply shall be visually examined, using a 10 power magnifier. The power supply shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.7) as evidenced by growth colonization (see 6.7) which includes branching (see 6.8.1) and sporulation (see 6.7) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.7) shall not be included in this evaluation.

3.8.8 Rain. After testing as specified in 4.8.10, the power supply shall meet full specification performance for the following measurements:

Paragraph	Measurements
3.7.2	4.7.1
3.7.3	4.7.2
3.7.4	4.7.3

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3.8.9 Dust. The power supply shall withstand blowing sand and dust as encountered in the desert. The power supply shall be in the operating configuration during the test of 4.8.11. After testing as specified in 4.8.11, the power supply shall meet full specification performance for the following measurement:

Paragraph	Measurement
3.7.2	4.7.1.1

3.8.10 Salt fog. After testing as specified in 4.8.12, the power supply shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to fluky, pitted, blistered, or otherwise loosened finish or metal surface.

3.8.11 Explosive atmosphere. The power supply shall withstand an explosive atmosphere without creating an explosion when tested in accordance with 4.8.13.

3.8.12 Bench handling. After testing as specified in 4.8.14, the power supply shall meet full specification performance for the following measurement:

Paragraph	Measurement
3.7.2	4.7.1.1

3.9 Electromagnetic compatibility (See 4.11). The power supply, when operated and loaded as specified herein, shall comply with the emission and susceptibility requirements of MIL-STD-461A, Notice 4 as specified below:

CE01(1)	CS01(1)	RE02
CE02(2)	CS02(3)	RE02.1(4)
CE04(2)	CS06(3)	

(1) Test to be performed on DC output leads only.
(2) Test to be performed on AC input leads and DC output leads.
(3) Test to be performed on AC input leads only.
(4) Upper frequency limit of test shall be 1000 MHz.

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3.9.1 Bonds, grounds, connectors and cables. Bonding for equipment current path return and shock hazards shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. Case or chassis mounted connectors shall have an approved RF conductive finish and shall be installed as above. Bonding impedance of the connector shell to the case shall not exceed 2.5 milliohms. Case mounted switches and meters shall be installed as above. Bonding impedance measured from the switch or meter body to the case shall not exceed 2.5 milliohms. Cables requiring overall shields shall have a minimum of 40 dB attenuation to electric fields over the frequency range of 10KHz to 400MHz. The shield shall be constructed of tinned copper braid. Connectors used with cables having an overall shield shall have an approved RF conductive finish and be provided with EMI backshells for peripheral bonding of the cable shield.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.9). Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.11 Reliability. The power supply shall have a specified MTBF (θ_0) of 1000 hours when operated under the following conditions:

a. Ambient temperature of 50°C (+5°C; - 0°C).

b. Duty cycle-continuous operation.

c. Environment-ground mobile.

The above requirement shall be demonstrated in accordance with 4.12.

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3.12 Systems safety.

3.12.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection. (See 4.10)

3.12.2 Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm). Compliance will be verified through a visual inspection. (See 4.10)

3.12.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses). Compliance will be verified through a visual inspection. (See 4.10)

3.13 Preconditioning.

3.13.1 Bounce preconditioning. After subjection to the bounce preconditioning, as specified in 4.6.1, the power supply shall be capable of meeting the requirements herein without subsequent processing.

3.13.2 Burn-in. Each power supply shall be subjected to a burn-in period as specified in 4.6.2.

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3.14 Workmanship (See 4.10). The equipment shall be manufactured and assembled in accordance with **requirement** 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268:

1. General requirements for plastic material and plastic parts.

2. Wiring and cabling, including: Slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

3. Flux and cleaning agents for solder.

4. Process for solder.

5. Cleaning.

6. Riveting.

7. General requirements for securing of parts.

8. Welding.

9. Tropicalization.

10. Mechanical operation of controls.

11. Self-tapping screws.

b. In this specification:

1. 3.5 Marking.

2. 3.6 Finish, protective.

3. 3.4 Printed wiring boards.

4. 3.2 Construction.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging. (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

1. Quality conformance inspection of equipment before packaging. (See 4.5)

2. Quality conformance inspection of packaging. (See 4.13)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The contractor shall furnish three (3) first article units of the complete Power Supply PP-6148( )/U.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table II, and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before packaging:

Item	Where required
Parts, materials, and processes; general	3.3
Printed wiring boards and protective coating	3.4, 3.4.1
Marking	3.5
Finish, protective	3.6

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TABLE II First Article Inspection

	INSPECTIONS	REQ. PARA.	TEST PARA.	UNIT 1	UNIT 2	UNIT 3
1.	Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units.		
2.	Preconditioning, Bounce Burn-in	3.13.1 3.13.2	4.6.1 4.6.2	Inspections to be performed on all units.		
3.	Group A Inspection	See Table III		Inspection to be performed on all units.		
4.	Group B Inspection	See Table IV		Inspection to be performed on all units		
5.	Environmental					
	Low Temperature Test	3.8.1.2	4.8.1.2	1		
	High Temperature Test	3.8.1.1	4.8.1.1	1		
	Temperature Co-efficient	3.7.11	4.7.9	2		
	Overload Protection	3.7.5	4.7.4	3		
	Current Limiting	3.7.7	4.7.4	4		
	Rain	3.8.8	4.8.10	5		
	Humidity	3.8.2	4.8.2	6		
	Altitude	3.8.3	4.8.3		1	
	Orientation	3.8.4	4.8.4		2	
	Electromagnetic Compatibility	3.9	4.11		3	
	Stability	3.7.10	4.7.8		4	
	Salt Fog	3.8.10	4.8.12		5	
	Explosive Atmosphere	3.8.11	4.8.13			1
	Dust	3.8.9	4.8.11			2
	Vibration	3.8.6.1	4.8.6			3
	Bounce	3.8.6.2	4.8.7			5
	Shock (Drop)	3.8.6.3	4.8.8			6
	Fungus	3.8.7	4.8.9			7
	Bench Handling	3.8.12	4.8.14			4
	Leakage (Immersion)	3.8.5	4.8.5		(See Note)	

NOTE: The leakage test shall be performed prior to and following vibration and shock testing of the same test unit.

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4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4, 4.6 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table III.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

Table III - Group A inspection

INSPECTION	REQ. PARA.	INSP. PARA.	AQL Major	dphu* Minor
Visual and mechanical	3.12	4.10	1.0dphu*	4.0dphu*
Input voltage	3.7.1	4.7.1	1.0% for the group	**
Output voltage	3.7.2	4.7.1 & 4.7.1.1		
Ripple voltage	3.7.3	4.7.2		
Efficiency	3.7.9	4.7.7	1.0%	
Battery discharge	3.7.8.1	4.7.6	1.0%	
Protection, general	3.7.8.2	4.7.4, 4.7.6 & 4.7.10	1.0%	

*dphu - Defects per hundred units.
**All electrical defects are major.

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4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQL's listed in Table IV shall be as follows:

AQL	INSPECTION LEVEL
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

Table IV - Group B inspection

INSPECTION	REQ. PARA.	INSP. PARA.	AQL
Recovery time	3.7.4	4.7.3	4.0%
Interchangeability	3.10	4.9	6.5%
Test for Battery Charging	3.7.6	4.7.5	4.0%

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4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

Table V - Group C inspection

INSPECTION	REQ. PARA.	INSP. PARA.
Group 1		
Leakage (Immersion)	3.8.5	4.8.5
Overload	3.7.5	4.7.4
Current Limiting	3.7.7	4.7.4
Group 2		
Stability	3.7.10	4.7.8
Orientation	3.8.4	4.8.4
Low Temperature Test	3.8.1.2	4.8.1.2
High Temperature Test	3.8.1.1	4.8.1.1
Temperature Coefficient	3.7.11	4.7.9
Group 3		
Altitude	3.8.3	4.8.3
Dust	3.8.9	4.8.11
Rain	3.8.8	4.8.10
Humidity	3.8.2	4.8.2
Vibration	3.8.6.1	4.8.6
Bench Handling	3.8.12	4.8.14
Bounce	3.8.6.2	4.8.7
Electromagnetic Compatibility	3.9	4.11
Salt Fog	3.8.10	4.8.12
Group 4		
Explosive Atmosphere	3.8.11	4.8.13
Shock (Drop)	3.8.6.3	4.8.8
Fungus*	3.8.7	4.8.9

*The power supply shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Sampling for inspection (Group 1). One sample shall be selected at random, from every 50 equipments or fraction thereof produced and subjected to each inspection specified in Table V. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.2 Sampling for inspection (Group 2). Two samples shall be selected at random each month and subjected to each inspection specified in Table V. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.3 Sampling for inspection (Group 3). One sample shall be selected at random, from each 200 equipments produced or fraction thereof, and subjected to each inspection specified in Table V. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. An additional unit shall be selected for EMI tests only. (See 4.5.3)

4.5.3.1.4 Sampling for inspection (Group 4). This inspection shall be required only when First Article samples are not specified by contract or purchase order. When required, one sample shall be selected at random, at the start of the contract, from the first quality conformance inspection production lot (see 4.5.3), and subjected to each inspection specified in Table V.

4.5.3.2 Order of inspection within Group C. The Group C inspections shall be performed in the order given in Table V. (See 6.5)

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4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.3.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.4 Group D inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

Table VI - Group D inspection

INSPECTION	REQ. PARA.	INSPEC PARA.
Reliability	3.11	4.12

4.5.4.1 Group D failures. In the case of a Group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed. (See 6.5)

4.5.4.2 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and passed Group A inspection.

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4.6 Preconditioning.

4.6.1 Bounce preconditioning. The unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick-strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After the bounce preconditioning, the unit shall not be repaired, alined, or otherwise changed prior to subjection to acceptance inspection. (See 3.13.1)

4.6.2 Burn-in. Each power supply submitted for acceptance shall be subject to a burn-in period with 24 hours of continuous failure free on time. The test shall be conducted at standard ambient temperatures (25° $+5^{\circ}$C) using an input of 115 vac, 60 Hz and an output of 28 vdc and 10 amperes. If a failure occurs, the equipment shall be repaired and returned to testing until it has accumulated 24 hours of continuous on time without a relevant failure. All failures shall be recorded and analyzed. A relevant failure for burn-in purposes shall be defined as the inability of the equipment to meet the requirements of this specification. (See 3.13.2)

4.7 Electrical tests.

4.7.1 Voltage tests, input and output. The output voltage shall meet the requirements of 3.7.2 when the load, input voltage, and frequency is varied as specified below. The input voltage and output voltage shall be measured at the terminals of the connectors:

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Input voltage (ac)	Freq. (Hz)	Resistive Load (amps)
103.5	50, 60, 400	0, 5, 10
115		0, 5, 10
126.5		0, 5, 10
207		0, 5, 10
230		0, 5, 10
253		0, 5, 10

4.7.1.1 Voltage adjustment. The output voltage of the power supply shall meet the voltage adjustment requirement of 3.7.2 with the power supply operating from a 115 vac, 60 Hz source, and the output current at 10 amperes (resistive load). The operating temperature shall be 25°C +5°C. The output voltage shall be measured at the terminals of the output connector.

4.7.2 Ripple voltage (See 3.7.3). The rms ripple voltage in the dc output shall be measured with an electronic voltmeter (Ballantine Model 300 or equal). The peak to peak ripple voltage shall be measured with a calibrated oscilloscope having a bandwidth of 20 Hz to 10 MHz (Tektronix Model 564 with 3A1 amplifier unit, or equal). This test shall be made using an input voltage of 115 Vac - 60 Hz, nominal output voltage of 28.0 Vdc, 10 ampere load current.

4.7.3 Recovery time. With the power supply operating from 115 vac - 60 Hz, the load (resistive) shall be switched from 10 to 0 amperes. The power supply shall be switched on and off at each of the load conditions. The voltage transients shall be noted on a calibrated oscilloscope. This procedure shall be repeated when the load is switched from 0 to 10 amperes. The transient voltage shall meet the requirements of 3.7.4. The output voltage shall be measured at the terminal of the output connector.

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4.7.4 Overload protection and circuit limiting. With the power supply operating at full load (10 amps) the load shall be increased gradually to a short circuit condition and the output current noted during the current limiting condition. Hold the short circuit condition for 10 seconds. Upon removal of the short, the power supply shall automatically return to normal operating conditions. Repeat five (5) times. The power supply shall then meet the requirements of 3.7.1, 3.7.2 and 3.7.3. This test to be performed at standard ambient temperatures (25° +5°C) and during the temperature tests of 4.8.1. the input voltage during the tests shall be 115 vac (rms), 60 Hz. Visual examination shall be made for any physical damage that might have occurred due to the overload condition. (See 3.7.5, 3.7.7 and 3.7.8.2)

4.7.5 Test for battery charging.

4.7.5.1 Vented nickel cadmium batteries. With the battery charger operating from 115 vac, 60 Hz, a fully discharged (20.0 volts) 5.5 ampere hour, 24 volt vented nickel cadmium secondary battery shall be used as a load. The output voltage of the battery charger shall be adjusted to 29.5 volts and current limiting set at 1.0 amperes. Continue the charge cycle for a period of eight (8) hours or until a battery voltage of 29.0 volts is reached. A 10 ohm resistor shall be used to discharge the battery to 20.0 volts prior to the test. (See 3.7.6.1)

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4.7.6 Battery discharge. With the input power source removed from the power supply, a fully charged 24 volt secondary type battery shall be connected across the dc output in correct polarity. An external ammeter shall be connected in the positive lead between the battery and the battery charger to measure the discharge (reverse) current. The battery shall then be connected in reverse polarity. The reverse current flow from the battery for either polarity shall not exceed ±50 milliamperes. (See 3.7.8.1 and 3.7.8.2.b)

4.7.7 Efficiency. The overall efficiency of the power supply shall be measured under conditions of full load, 28.0 vdc at 10 amperes, and at nominal 115/230 vac input, and at power frequencies of 50, 60 and 400 Hz. The percent efficiency shall be determined by the following and shall meet the requirements of 3.7.9:

$$\frac{\text{Output power (volts x current)}}{\text{Input power (measured with wattmeter}} \times 100 = \% \text{ eff.}$$

4.7.8 Stability. The power supply shall be operated from a 115 vac, 60 Hz source with a 10 ampere load at nominal 28 vdc output for 24 hours at standard ambient temperature (25° ±5°C). At the end of this test, the power supply shall meet the requirements of 3.7.10. Repeat this test using an input of 115 vac, 400 Hz.

4.7.9 Temperature coefficient. The power supply shall be operated from 115 V 60 Hz source with a 10 ampere load at nominal 28 vdc output for 4 hours in an ambient temperature of 21° ±2°C. The ambient temperature shall then be increased to 49° ±2°C and the equipment operated for 4 hours. At the end of this test, the change in the output voltage shall not have exceeded 1.0 percent. Repeat this test at an ambient temperature starting at -40° ±2°C and increasing to -12° ±2°C. (See 3.7.11)

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4.7.10 General protection (See 3.7.8.2). With the AC input adjusted to 115 vac, 60 Hz, adjust the DC output to 28 vdc with a 10 ampere load. Raise the input voltage to 139 vac and then reduce it back to 115 vac. Reduce the input voltage to 92 vac and then raise it back to 115 vac. At the end of the test, the power supply shall conform to the requirements of 3.7.2, 3.7.3 and 3.7.7.

4.8 Service conditions (See 3.8).

4.8.1 Temperature.

4.8.1.1 High temperature. The power supply shall be subjected to the test of Method 501, Procedure II, of MIL-STD-810. The internal chamber temperature in Step 4 shall be 68.3°C. For Step 7, the internal chamber temperature shall be adjusted to 68.3°C and maintained until temperature stabilization of the test item is reached. Operate the power supply with a 10 ampere load at nominal 28 volts output until stabilization of the test item is reached. At the conclusion of Step 7, the power supply shall meet full specification performance of the measurements specified in 3.8.1.1.

Note: To determine when the power supply has stabilized, thermocouples shall be placed on critical components and those having the longest thermal lag; these shall include switching transistors, bridge rectifiers and regulator transistors. Operation at the high temperature extreme not to exceed 4 hours, at any one time.

4.8.1.2 Low temperature. The power supply shall be subjected to the test of Method 502, Procedure I, of MIL-STD-810. For Step 4, the internal chamber temperature shall be adjusted to -45.6°C and maintained until temperature stabilization of the test item is reached. During and after testing, the power supply shall meet full specification performance of the measurements specified in 3.8.1.2.

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NOTE: To determine when the power supply has stabilized, thermocouples shall be placed on critical components and those having the longest thermal lag; these shall include switching transistors, bridge rectifiers and regulator transistors. Operation at the low temperature extreme not to exceed 72 hours, at any one time.

4.8.2 Humidity. The power supply in its normal operating configuration, shall be subjected to the test of Method 507, Procedure III of MIL-STD-810, and shall meet the requirement of 3.8.2 when tested as specified. In Step 6 the power supply shall be operated during the last five hours of the last cycle. In Step 8 the power supply shall be operated during the last five hours of the steady state exposure.

4.8.3 Altitude. The power supply in its normal operating configuration shall be subjected to the test of Method 500, Procedure I of MIL-STD-810, except that operation of the equipment is required at a pressure of 523 mm of Hg (20.6 inches of mercury or 10,000 feet above sea level). The power supply shall be exposed for one hour to a pressure of 87.5 mm Hg (3.44 inches of mercury or 50,000 feet above sea level, non-operating). (See 3.8.3)

4.8.4 Orientation. The power supply shall be inclined up to 90° (plus 3, minus 1 degree) for a minimum of 5 minutes in each plane (forward, backward, left, and right). While in each position perform the test of 4.7.1.1. (See 3.8.4)

4.8.5 Leakage (immersion). The power supply, with case closed, top cover installed, shall be subjected to the test of Method 512, Procedure 1 of MIL-STD-810. (See 3.8.5)

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4.8.6 Vibration. Vibration of the power supply shall be measured as follows, to determine conformance to 3.8.6.1:

a. Secure the power supply directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

b. Vibrate the power supply seccessively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

c. In each of three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

d. Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.8.7 Bounce. The power supply shall be tested on package tester type 1000-SC as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The test shall be as follows:

a. Cover the tester bed with a panel of 1/2 inch plywood, with the grain parellel to the drive chain. Space six-penny nails, with heads below the surface at 6 inch intervals around its four edges and at 3 inch intervals in a 6 inch square at the center of the test area.

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b. Using suitable wooden fences, constrain the test item to a horizontal motion of not more than 2 inches in a direction parallel to the axes of the shafts of the package tester, and in a direction perpendicular to the axes of the shafts, a distance more than sufficient to insure the test item will not rebound from fence to fence.

c. The test item as prepared for field transportation, shall be placed on the package tester within the constraints outlined above. The package shall be operated in the synchronous mode with the shafts in phase. (In this mode any point on the bed of the tester will move in a circular path in a vertical plane perpendicular to the axes of the shafts).

d. Operate the package tester, shafts in phase, for a total of 3 hours at 283 ±1 rpm. Turn the test item to rest on a different face at the end of each 1/2 hour period, so that at the end of the 3 hour period the test item will have rested on each of its six faces (top, bottom, sides, ends).

e. Upon completion of the test, the power supply shall be capable of meeting the requirements of 3.8.6.2.

4.9.8 Shock (drop). The power supply as prepared for field use shall be subjected to the test of Method 516, Procedure II of MIL-STD-810. Drops shall be made from a quick-release hook or drop tester. (See 3.8.6.3)

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4.8.9 Fungus. The power supply shall be subjected to the fungus test of Method 508, Procedure I of MIL-STD-810. There shall be abundant growth colonization (6.7.3) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the power supply is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the power supply to meet the requirements of 3.8.7 shall constitute failure of this test. Operation of the power supply is not required during the test.

4.8.10 Rain. The power supply in its normal operating configuration shall be subjected to the test of Method 506, Procedure I of MIL-STD-810. Perform the test of 4.7.1.1 during the last 10 minutes of the 30 minute rainfall. Complete the final measurements within one hour after the rain has been discontinued. (See 3.8.8)

4.8.11 Dust. The power supply in its normal operating configuration shall be subjected to the test of Method 510, Procedure I of MIL-STD-810. (See 3.8.9)

4.8.12 Salt fog. The power supply shall be subjected to the test of Method 509, Procedure I of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flackey, pitted, blistered or otherwise loosened finish on metal surface. (See 3.8.10)

4.8.13 Explosive atmosphere. The power supply shall be subjected to the test of Method 511, Procedure I of MIL-STD-810. Operation of the power supply is required at maximum altitude of 10,000 feet. (See 3.8.11)

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4.8.14 Bench handling. The power supply shall be subjected to the test of Method 516, Procedure V of MIL-STD-810. (See 3.8.12)

4.9 Interchangability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable subassemblies and replaceable parts, when such dimensions affect mating of parts.

b. Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

c. Location of fasteners on separate parts or assemblies which must mate together such as cases, covers, and mountings.

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4.10 Visual and mechanical inspection. The power supply shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of 3.12 (Systems Safety which can be determined visually), and 3.14.

4.11 Electromagnetic compatibility. The power supply shall be tested for compliance to the requirements of 3.9. During the test, the power supply shall be operated from a 115 vac, 60 Hz source with a 6 ampere load at nominal 28 vdc. Test setups and measurement techniques shall be in accordance with MIL-STD-462, Notice 3, as detailed in a contractor prepared Government approved test plan. Interconnecting cables used during test shall be similar in type, shielding length and connectors, to those to be used in an operational installation.

4.11.1 Bonds, grounds, connectors and cables. Prior to the performance of EMI tests, the bonding impedance requirements of 3.9.1 shall be verified. Bonding impedance of the following items shall be measured with a Shallcross Meter Model 670A or equal:

a. All connector shells to chassis case of frame.

b. All switches, meters, hinge fasteners, transformers and iron core inductors.

c. Power filter body to ground and chassis frame.

d. All surfaces requiring EMI gaskets.

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4.12 Reliability testing. (See 3.11)

4.12.1 Reliability qualification test. The contractor shall conduct a reliability qualification test on the first ten units produced in accordance with the test conditions of 4.12.3.

4.12.2 Reliability acceptance test. The contractor shall conduct a reliability acceptance test on 10 units from each lot based on the test conditions of 4.12.3. A lot for reliability purposes shall be defined as either 60 units or one month's production whichever is larger.

4.12.3 Reliability test conditions. The reliability test shall be based on a MTBF of 1000 hours. The test shall be run according to the conditions of test Level C of MIL-STD-781, equipment cycling shall be limited to 4 hours of on time per cycle. Accept-reject criteria shall be determined using test plan IV of MIL-STD-781. Each unit shall have an input of 115 vac at 60 Hz and an output of 28 vdc and 10 amperes. A failure for reliability purposes shall be defined as the inability of any equipment to meet the requirements of this specification. Each unit shall be continuously monitored to detect any failures. The total test time for each equipment shall be in accordance with 4.5.2.4 of MIL-STD-781. 5.5 thru 5.9 of MIL-STD-781 shall apply.

4.13 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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4.13.1 Rough handling test (packaging). When the rough handling test in accordance with MIL-P-116 is required by contract (see 6.2.f) the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Insp.Para.	Measurement	Input Voltage
4.7.1	Output voltage (no load and full load)	115 vac rms, 60 Hz

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A, B or Commercial as specified (see 6.2.b)

5.1.1 Level A.

5.1.1.1 Cleaning. Each power supply shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each power supply shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. Not applicable

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be individually unit packed method IC-1.

5.1.1.4.2 Cable assembly, battery charging. Each battery charging cable assembly shall be individually unit packed in accordance with MIL-C-55442.

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5.1.1.4.3 Power Supply PP-6148( )/U. Each power supply shall be individually unit packed method III as follows: Secure power cable assembly in case cover and fasten cover to case body. The power supply, together with the battery charging cable assembly unit packed as specified in 5.1.1.4.2, shall be cushioned on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather-resistant, variety SW, Grade W5c, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be inclosed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2.b). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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5.2.1 Level A. A quantity of power supplies, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, over-seas type; PPP-B-621, Style 4, Class 2; or PPP-B-585, Style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type I, Finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load Type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, Type IV, Group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A, or shrink film.

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5.2.2 Level B. A quantity of power supplies, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, Class 2, Style E, or PPP-B-636, Type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, Group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of power supplies, unit packed as specified in 5.1, shall be placed on a pallet, load Type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, Type IV, Group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. Power Supply PP-6148( )/U is intended to provide dc power for operation of Radio Set AN/PRC-70 and charge nickel cadmium secondary batteries from a 115/230 volt, 50, 60, or 400 Hz single phase ac power source.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A, Level B or Commercial preservation and packing. (see Section 5)

c. First article inspection:

1. Three sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit. (See 3.1).

2. Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.
(See 4.13.1)

d. Marking and shipping of samples.

e. Place of final inspection.

f. When rough handling test is required. (See 4.13.1)

g. Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer. (See 3.14)

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6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-6148W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Group C and D inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2 and 4.5.4.1)

6.6 Color. The color chip furnished by the procuring agency (3.6) will match color chip No. X-24087 and may be obtained upon request to Commanding Officer, U.S. Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pa., 19120.

6.7 Definitions.

6.7.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by sheets or secondary stems growing from the main stem or filament (hypha).

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6.7.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitues abundant growth.

6.7.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.7.4 Sporulation. Sporulation is the formation of minute unicelluar reproductive or dormant bodies, called spores.

6.7.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungla filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.8 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specification or preparing activity for recommended disposability methods.

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NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
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